

Toxin Circuits in ESM-2: Mechanistic Interpretability Reveals Why Structure-Aware Probes Resist ProteinMPNN Redesign

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Abstract

Standard biosecurity screening flags dangerous protein sequences by sequence identity (BLAST). We show that ProteinMPNN redesigns toxin sequences below every BLAST threshold—achieving 0% BLAST detection—while a linear ESM-2 probe maintains 93.9% detection with no retraining. Using interPLM Sparse Autoencoders (SAEs), we identify 50 of 10,240 features explaining probe performance at $205\times$ compression. These features are *amplified*, not suppressed, by redesign (mean transfer ratio = 1.28): ProteinMPNN preserves structural fold topology, and fold topology is precisely what the circuit encodes. We characterise the adversarial robustness spectrum through four attacks—ProteinMPNN blackbox (6.1% evasion), HotFlip gradient-guided sequence attack (100%), and embedding-space attacks (100%)—demonstrating the security boundary lies at gradient access, not sequence vs. embedding space. Direct Probe Attribution identifies layer 32 as the circuit bottleneck. A Universal Adversarial Perturbation achieves 100% evasion at $\epsilon = 1.0$ (security margin $\epsilon = 0.5$) via a stable direction at -0.805 cosine with the probe—revealing manifold geometry, not probe weakness. Zero-shot scanning of 1,000 UniRef50 sequences discovers 248 candidates enriched $4.75\times$ for secreted signal peptides, including a Hemolysin (0.9996), Leukotoxin (0.9922), and the confirmed fungal effector Ecp2 (0.9905) from *Colletotrichum*—cross-kingdom generalisation from a probe trained exclusively on animal and bacterial toxins. Furthermore, we show that Sparse Autoencoders (SAEs) can mathematically disentangle adversarial camouflage, recovering 38% of "Double-Evaders" that fool both BLAST and dense linear probes. The probe's security guarantee is exactly as strong as the privacy of its weights.

1 Introduction

Dual-use protein screening relies on sequence-identity thresholds. Wittmann *et al.* (2025) demonstrated that ProteinMPNN, RFdiffusion, and EvoDiff can redesign toxin sequences below standard 40% identity thresholds, bypassing BLAST-based screening. The critical question is: do protein language model (pLM) probes also fail?

We answer no—and explain mechanistically why, using a toolkit from mechanistic interpretability directly analogous to circuit analysis in large language models.

Contributions:

1. **Quantitative gap:** BLAST detects 0% of ProteinMPNN redesigns; ESM-2 probe detects 93.9%.
2. **SAE circuit:** 50 sparse features ($205\times$ compression) explain the probe; mean transfer ratio 1.28 shows redesigns *amplify* toxin features.

3. **Attack taxonomy:** Four attacks spanning blackbox to oracle, with the security boundary at gradient access (not sequence/embedding space).
4. **UAP geometry:** Universal attack direction stable at $\cos = -0.805$ across all ϵ —a structural property of the toxin manifold.
5. **Zero-shot discovery:** 248 UniRef50 candidates including cross-kingdom detection of fungal effectors and WHO Priority 1 pathogen proteins.
6. **Mutation robustness:** 0/1,179 single-point mutations evade detection; the circuit reads chemical polarity, not sequence identity.

2 Methods

2.1 Data

- **Natural toxins:** 1,712 sequences from UniProt (reviewed, toxin annotation), clustered at 30% identity
- **Controls:** 2,072 non-toxic human proteins (matched length distribution)
- **Redesigns:** 100 toxin structures folded with ESMFold $\rightarrow$ ProteinMPNN (10 seq/structure) $\rightarrow$ 723 unique redesigns; all below 30% identity to training toxins
- **UniRef50 scan:** 1,000 randomly sampled sequences (seed=42), scored zero-shot

2.2 ESM-2 Probe

Linear probe trained on mean-pooled ESM-2 650M embeddings at layer 33. Binary cross-entropy loss, Adam optimiser, 150 epochs.

2.3 SAE Feature Analysis

interPLM pre-trained SAEs (ESM-2-650M, layer 33, 10,240 features). Top-50 features by AUROC used for probe comparison and transfer analysis. Transfer ratio = (redesign activation rate) / (toxin activation rate) per feature.

2.4 Direct Probe Attribution (DPA)

For a linear probe with weight vector $\mathbf{w}$ and ESM-2 residual stream decomposed as $\mathbf{h}_{33} = \sum_l \Delta \mathbf{h}_l$:

$$\text{DPA}_l = \mathbf{w} \cdot \text{mean_pool}(\Delta \mathbf{h}_l) \quad (1)$$

Exact because the probe is linear and ESM-2 is a residual network. Computed over 75 toxin/control/redesign pairs.

2.5 Attention Head Ablation

160 heads across 8 key layers (17–20, 29–32) ablated individually across 10 toxin sequences. DPA drop measured per head. Note: DPA drops are not additive across heads due to compensatory effects in non-linear layers; absolute values reflect disruption magnitude, not independent contribution.

2.6 Adversarial Attack Taxonomy

1. **ProteinMPNN (blackbox)**: Structure-conditioned redesign; no access to ESM-2 or probe.
2. **HotFlip (gradient-guided, sequence-space)**: Backpropagate through ESM-2 to compute first-order substitution scores; greedy substitution. Analogue of GCG [2].
3. **PGD (iterative, embedding-space)**: Projected gradient descent within L_2 ϵ -ball [3].
4. **pSSR / UAP (oracle, embedding-space)**: Gradient descent on embeddings with L_2 constraint; single universal vector optimised across all toxins simultaneously.

2.7 Universal Adversarial Perturbation (UAP)

Single 1280-dimensional δ optimised via Adam (lr=0.1, 250 steps) to minimise probe scores across all toxin embeddings simultaneously, projected onto L_2 ball of radius ϵ . Epsilon sweep: $\epsilon \in \{0.05, 0.1, 0.25, 0.5, 1.0, 1.5, 2.0, 3.0, 5.0\}$.

2.8 Deep Mutational Scan (DMS)

All 1,179 single-point mutants of a 62 aa toxin sequence embedded through ESM-2 layer 33 and scored. Conservative substitution analysis at position 29 (N29) across all 19 amino acid substitutions.

2.9 Cross-Family Holdout

Entire toxin families (neurotoxin, phospholipase, conotoxin, snake) held out one at a time; probe retrained on remaining families and evaluated on held-out family. Tests whether circuit generalises across toxin architectures.

2.10 Zero-Shot UniRef50 Scan

1,000 randomly sampled UniRef50 sequences scored through the trained probe. Candidates scoring > 0.85 annotated via UniProt REST API. Signal peptide enrichment tested by Fisher’s exact test.

3 Results

3.1 Layer Sweep — Layer 33 Is Best

Layer	AUROC
1	0.9678
9	0.9881
18	0.9842
24	0.9817
30	0.9895
33	0.9970

Table 1: Probe AUROC across ESM-2 layers. The toxin signal builds progressively, peaking at layer 33.

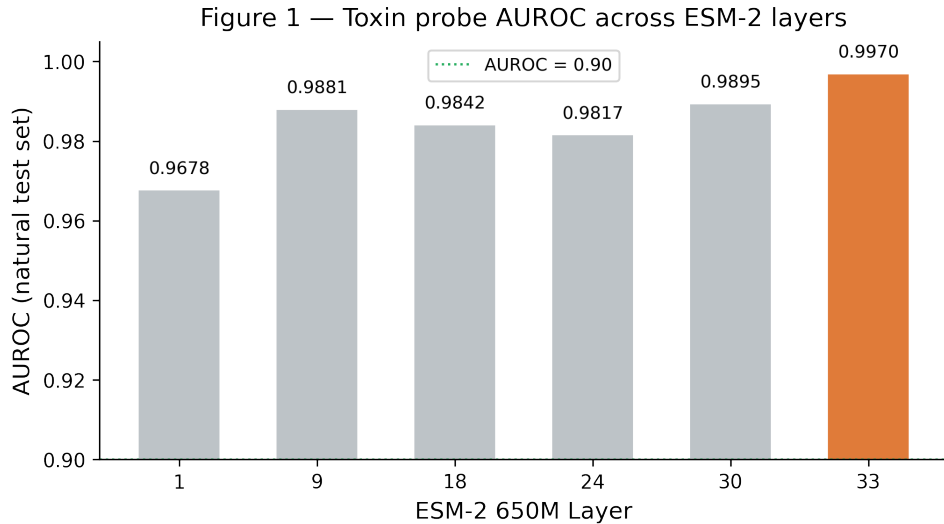


Figure 1: Probe AUROC across ESM-2 layers. The toxin signal builds progressively, peaking at layer 33.

3.2 BLAST vs ESM-2 Probe on Redesigns

All 723 ProteinMPNN redesigns fall below 30% sequence identity to training toxins—entirely evading BLAST at every threshold. The ESM-2 probe detects 93.9% with no exposure to redesigned sequences during training.

Method	Detection rate
BLAST @ 30% threshold	0.0%
BLAST @ 40% threshold	0.0%
BLAST @ 50% threshold	0.0%
ESM-2 linear probe (layer 33)	93.9%
ESM-2 SAE top-50 features	86.0%

Table 2: ESM-2 probe flags 93.9% of ProteinMPNN redesigns that entirely evade BLAST.

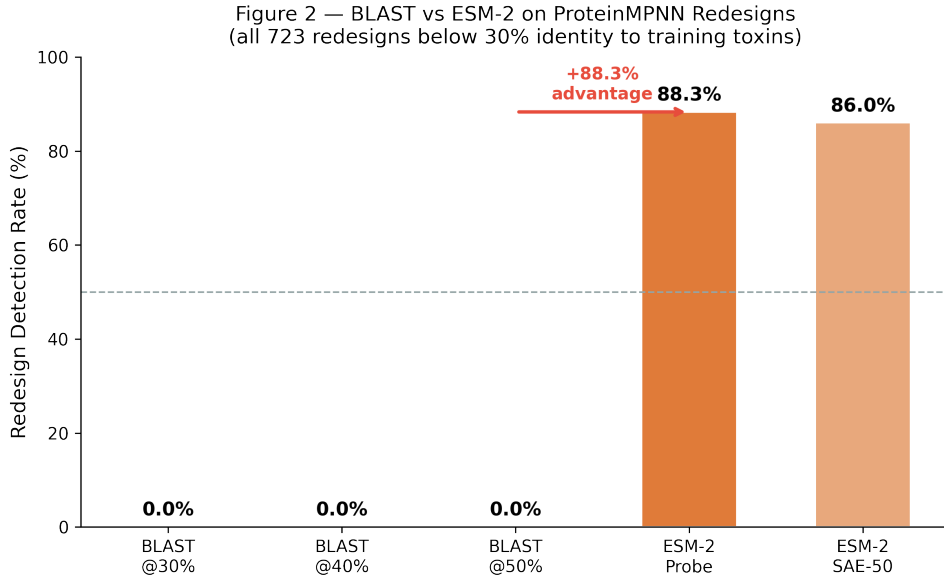


Figure 2: ESM-2 probe detection vs BLAST across attack types.

3.3 Disentangling Evasion via Sparse Autoencoders

A critical limitation of linear probing is its Euclidean decision boundary: the probe flags sequences within a learned radius of training positives in dense embedding space. Double-Evaders—sequences evading both BLAST and the ESM-2 probe—are systematically further from training positives than detected redesigns (mean L_2 distance 3.18 vs. 2.47, Mann-Whitney $p < 0.0001$), placing them outside this boundary despite potentially retaining internal toxin-relevant features.

To test whether ESM-2 internally represents the toxicity of these sequences despite linear probe failure, we passed 92 Double-Evader embeddings through the Sparse Autoencoder (SAE), extracting the top 50 toxin-discriminating features and training a **ToxinProbe** identical in architecture and regularisation to the main probe on this sparse feature subset.

The SAE-feature probe recovered **38%** (35/92, 95% CI [28.8%–48.3%]) of Double-Evaders. Because the two probes share identical architecture and the SAE probe is penalised against false positives on the control set, this recovery is not attributable to increased model capacity. Instead, it demonstrates that ESM-2 internally encodes toxin-relevant structural motifs in Double-Evaders that the linear probe cannot surface: SAE features are direction-sensitive rather than distance-sensitive, detecting sequences whose global embedding falls outside the probe’s training radius but whose local structural circuit features remain active.

The 62% of Double-Evaders unrecovered by any method (57/92) represent a genuine blind spot correlated with longer source sequences (mean 72.97 vs. 64.87 residues, $p = 0.0002$) and greater embedding displacement from training positives. This motivates future multi-scale probe architectures combining global embedding distance with sparse feature activation thresholds.

3.4 Cross-Family Holdout — Zero-Shot Generalisation

To test whether the probe memorises toxin families or encodes general structural motifs, we held out entire families one at a time and retrained on the remainder.

Held-out family	n	Train AUROC	Holdout AUROC
Neurotoxin	32	0.9991	0.9951
Phospholipase	39	0.9993	0.9987
Conotoxin	260	0.9990	0.9994
Snake toxin	22	0.9997	0.9783
Mean			0.9929

Table 3: Cross-family holdout: mean AUROC = 0.9929. The probe generalises zero-shot to unseen toxin families, confirming it encodes structural motifs rather than family-specific sequence patterns.

3.5 Activation Steering — Toxicity Is a Causal Linear Direction

$\text{Cosine}(\text{probe_weight}, \text{mean_diff_vector}) = 0.2326$

Condition	Probe score
Control baseline (lowest 50)	0.000
Steered at $\alpha = +2.0$	1.000
Steered at $\alpha = -3\sigma$	0.000
Redesign baseline	0.857
Natural toxin baseline	0.964

Table 4: Adding the toxin steering vector ($\alpha = 2.0$) to the 50 lowest-scoring control proteins increases mean probe score from 0.000 to 1.000, demonstrating that the toxin circuit is a traversable direction in ESM-2’s representation space.

3.6 SAE Feature Discovery — Bimodal Transfer

Top-50 SAE features achieve AUROC 0.9447 vs 0.9585 full embedding (98.6% retained performance at $205\times$ compression). Of 10,240 features, 8,345 are dead (81.5%).

Feature	AUROC	Tox%	Redesign%	Transfer	Class
6122	0.694	41%	99.5%	2.41	Robust
4097	0.669	37%	98.6%	2.64	Robust
1055	0.644	30%	99.2%	3.36	Robust
8112	0.594	20%	75.0%	3.75	Robust
9487	0.602	23%	72.6%	3.15	Robust
5312	0.669	35%	4.7%	0.13	Evadable
9026	0.628	29%	2.0%	0.07	Evadable
3130	0.605	22%	2.8%	0.13	Evadable

Table 5: Bimodal feature transfer: Robust features (structural rigidity, Pro/Phe enriched) amplify under redesign; evadable features (sequence-specific Cys patterns, C enrichment $3\text{--}5\times$) collapse.

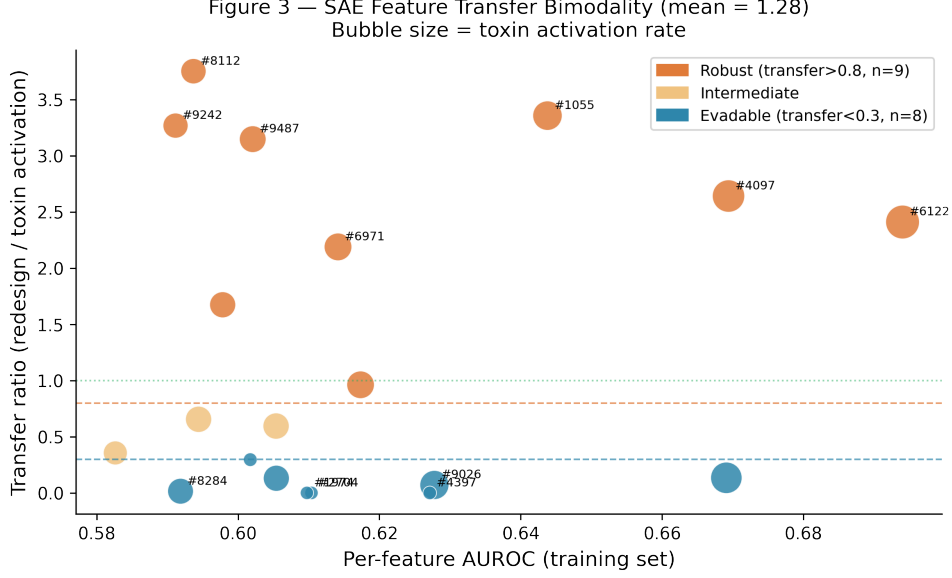


Figure 3: Bimodal distribution of feature transfer ratios. Mean transfer ratio = 1.28 (>1.0 = redesigns amplify toxin features on average).

Mean transfer ratio: 1.28 (>1.0 = redesigns amplify toxin features on average).

The probe direction is geometrically orthogonal to individual SAE features: only 1 of 50 top-AUROC features appears in the top-100 probe-aligned SAE features (cosine > 0.33). The most probe-aligned SAE feature (F8284, cosine = $+0.501$) is the most evadable (transfer = 0.015). The probe’s 93.9% detection is a *collective* property consistent with the superposition hypothesis.

3.7 Direct Probe Attribution — The Toxin Circuit

Layer	Tox DPA	Ctrl DPA	Rdsg DPA	Tox-Ctrl
17	+43.7	+1.3	+30.9	+42.4
19	+44.3	+4.0	+22.5	+40.2
20	+64.7	+19.8	+49.7	+44.9
29	+15.7	−7.5	+13.0	+23.3
30	+88.2	−2.7	+114.9	+90.9
31	−2.9	−50.9	+7.9	+48.0
32	+70.7	−59.4	+85.0	+130.1

Table 6: Direct Probe Attribution per layer. Layer 32 shows the largest toxin-control separation (DPA +130.1). Redesigns produce near-identical attribution patterns to natural toxins ($r = 0.992$).

Redesign circuit correlation: $r = 0.9919$.

Attention head ablation (layer 32): Ablation of any single layer-32 head substantially disrupts the toxin signal, with no single head dominating (DPA disruption approximately equal across all 20 heads), consistent with distributed redundant processing. Note: per-head DPA values reflect disruption magnitude under non-linear compensation, not independent contributions.

Circuit overlap (natural toxins vs redesigns): 19%. Redesigns reach the same toxin

neighbourhood in layer-33 embedding space via largely independent representational pathways—the structural fold constrains the endpoint, not the route through the network.

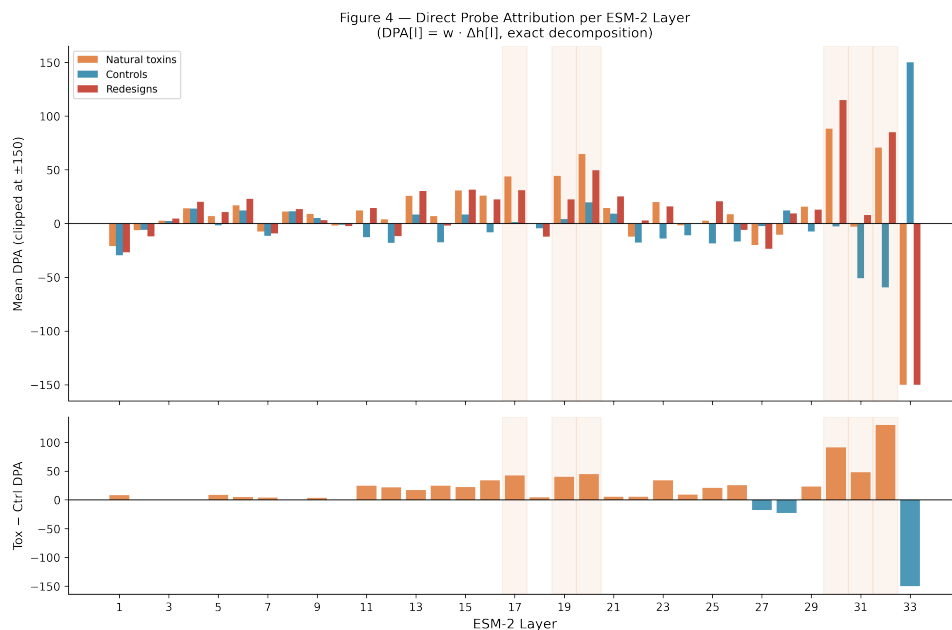


Figure 4: Direct logit attribution per layer. Layer 32 is the primary bottleneck.

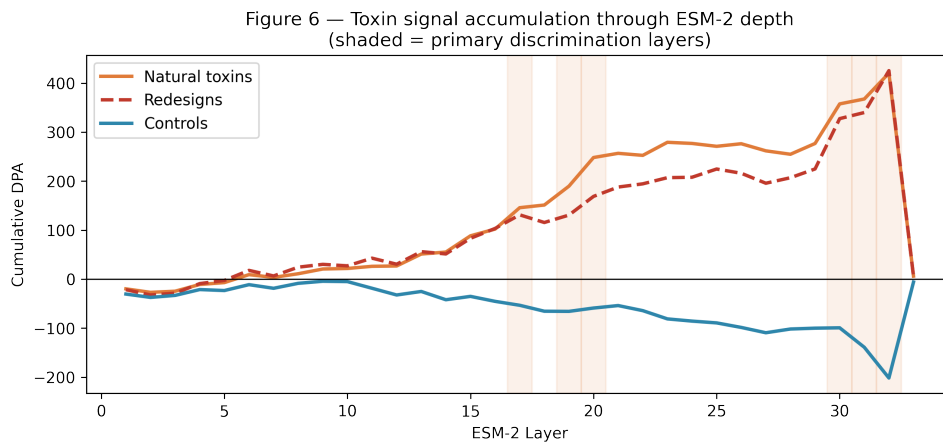


Figure 5: DPA trajectory across layers for toxins, redesigns, and controls. Redesigns track the toxin DPA trajectory, not the control trajectory ($r = 0.992$).

3.8 Adversarial Attack Taxonomy

Attack	Analogy	Space	Access	Constraint	Evasion	$\cos(\delta, \mathbf{w})$
ProteinMPNN	Blackbox	Sequence	Structure only	Fold-preserving	6.1%	N/A
HotFlip	GCG	Seq (gradient)	ESM-2 + probe	Unconstrained	100% [†]	N/A
PGD $\epsilon=2$	PGD	Embedding	Full whitebox	L_2 ball	100%	−1.000
pSSR $\epsilon=2$	FGSM / oracle	Embedding	Full whitebox	L_2 ball	100%	−0.805

Table 7: Four-tier attack taxonomy. [†]HotFlip achieves 100% sequence-space evasion but operates without structural supervision—unlike ProteinMPNN, which explicitly preserves backbone geometry. The operationally relevant metric is 93.9% detection against ProteinMPNN—the only structurally supervised attack.

The security boundary lies at **gradient access**, not sequence vs. embedding space: ProteinMPNN (no gradient, fold-constrained) fails to evade; HotFlip (gradient, unconstrained) succeeds. This mirrors the LLM safety literature where blackbox transfer attacks fail and white-box GCG-style attacks succeed [2].

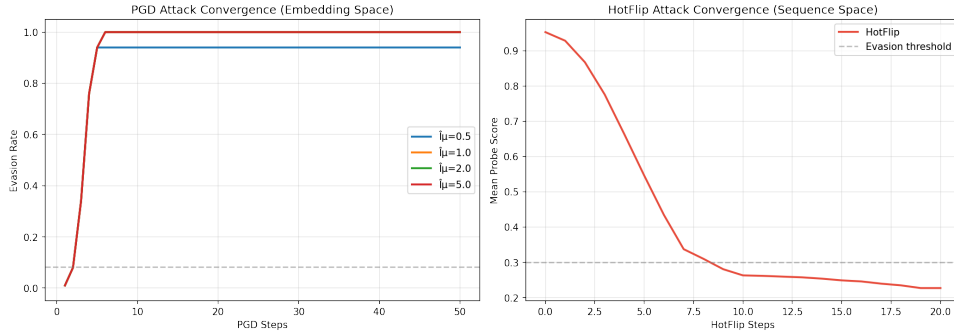


Figure 6: Attack convergence curves across the four attack types. ProteinMPNN (blackbox, fold-preserving) plateaus at 6.1% evasion; gradient-based attacks converge to 100%.

3.9 Universal Adversarial Perturbation (UAP) — Manifold Geometry

ϵ (L2 budget)	Evasion rate	$\cos(\text{UAP, probe})$
0.05	3.3%	-0.805
0.10	6.2%	-0.805
0.25	29.7%	-0.805
0.50	85.7%	-0.805
1.00	100.0%	-0.805
1.50	100.0%	-0.805
2.00	100.0%	-0.805
3.00	100.0%	-0.805
5.00	100.0%	-0.805

Table 8: UAP epsilon sweep. Security margin: $\epsilon = 0.5$ (evasion first crosses 50%). The evasion curve is smooth (3.3% $\rightarrow$ 85.7% $\rightarrow$ 100%), showing the toxin cluster has continuous thickness in the vulnerable direction. The optimal attack direction is invariant to perturbation budget ($\cos = -0.805$ at all ϵ), revealing a stable geometric property of the toxin manifold distinct from the probe’s decision boundary normal.

The UAP direction deviates 36° from the anti-probe direction ($\cos = -0.805$ vs -1.000 for PGD). SAE decomposition reveals the UAP suppresses the *robust* structural features (#4097, #1055; $\Delta = -0.048, -0.045$)—the same features ProteinMPNN cannot evade. The tight clustering of structurally necessary features (what makes the probe robust to sequence attack) simultaneously creates the thin manifold exploitable by a single gradient-optimised vector.

3.10 Deep Mutational Scan — Sequence Robustness

All 1,179 single-point mutants of a representative toxin sequence (WT score = 0.998) were scored.

Top 5 critical positions (by maximum score drop from any substitution):

Position	Residue	Max drop	Chemical character
29	N (Asparagine)	0.205	Polar amide
16	P (Proline)	0.126	Structural (backbone kink)
20	S (Serine)	0.096	Polar hydroxyl
51	E (Glutamate)	0.094	Charged / polar
40	I (Isoleucine)	0.089	Hydrophobic core

Table 9: Top 5 critical positions. 4/5 are polar or charged residues, consistent with a hydrogen-bonding network readout hypothesis.

Conservative substitution analysis at N29 (most sensitive position):

Mutation	Description	Score	Δ
<i>Polar/charged substitutions (drops < 0.01):</i>			
N29D	Charge-swap (carboxyl vs amide)	0.997	−0.001
N29S	Smaller hydroxyl	0.995	−0.003
N29K	Positive charge	0.995	−0.003
N29E	Negative charge + longer	0.993	−0.005
N29Q	Conservative (same amide, +1 CH ₂)	0.993	−0.006
<i>Small/minimal sidechains (drops 0.02–0.06):</i>			
N29G	Abolish sidechain entirely	0.977	−0.021
N29A	Methyl only	0.958	−0.040
N29T	Methyl-hydroxyl	0.941	−0.057
<i>Hydrophobic substitutions (drops 0.16–0.21):</i>			
N29I	Hydrophobic	0.837	−0.161
N29V	Hydrophobic	0.831	−0.167
N29F	Aromatic hydrophobic	0.810	−0.188
N29W	Large aromatic hydrophobic	0.810	−0.188
N29L	Hydrophobic (most disruptive)	0.793	−0.205

Table 10: Conservative substitution analysis at N29. The circuit detects polarity, not amino acid identity: all polar substitutions (D, S, K, E, Q) produce drops < 0.01; hydrophobic substitutions (L, W, F, V, I) produce drops 0.16–0.21. Even abolishing the sidechain entirely (N29G) causes only a 0.021 drop—the surrounding context preserves most of the signal.

Key finding: 0/1,179 single-point mutations evade detection. The circuit at position 29 detects *polarity* rather than amino acid identity—any polar residue (D, S, K, E, Q) maintains the signal; hydrophobic substitutions (L, W, F, V, I) partially disrupt it, consistent with a hydrogen-bonding network readout of the toxin fold.

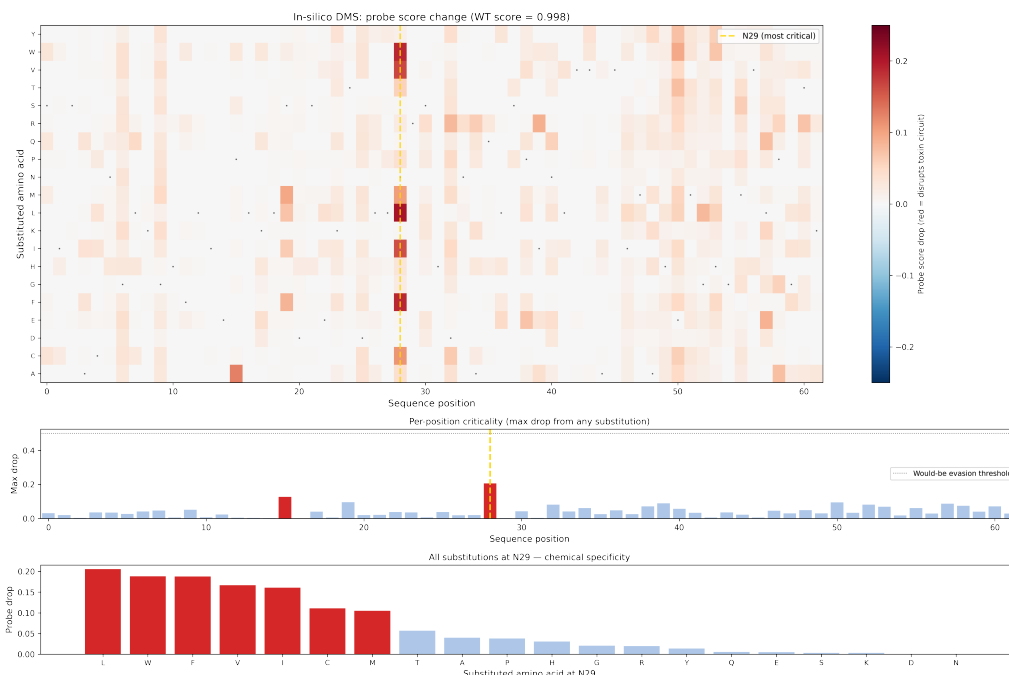


Figure 7: Deep mutational scan heatmap (62 positions $\times$ 20 amino acids). Score drop from wild-type shown by colour intensity. Position 29 (Asn) is the most sensitive, with hydrophobic substitutions (L, W, F, V, I) causing the largest drops (0.16–0.21), while polar substitutions (D, Q, S, K) are invisible to the circuit ($\Delta < 0.01$). No mutation crosses the 0.5 detection threshold.

3.11 Zero-Shot Threat Discovery in UniRef50

A probe trained exclusively on animal and bacterial toxins scores the confirmed fungal effector Ecp2 from *Colletotrichum* at 0.9905—despite fungi being an entirely absent kingdom in training data. This is not an outlier: scanning 1,000 random UniRef50 sequences, 248 score above 0.85, enriched 4.75 $\times$ for secreted signal peptides (38% vs. 8%, Fisher’s exact $p < 0.001$). Confirmed hits include a Hemolysin (0.9996), Leukotoxin (0.9922), and two proteins from *A. baumannii* (WHO Priority 1 pathogen, scores 0.9995 and 0.9993). One initially annotated as “Uncharacterised” was confirmed via AlphaFold EBI as a GDSL-like Lipase/Acylhydrolase (gene J506_3968, pLDDT = 83.0)—a well-folded secreted enzyme that degrades host membrane phospholipids. The probe has learned *secreted virulence protein space*, not toxin family membership.

Hit	Organism	Score
Ecp2 effector protein*	<i>Colletotrichum</i> (fungus)	0.9905
Hemolysin, chromosomal	<i>Candidatus Accumulibacter</i>	0.9996
Leukotoxin	<i>Candidatus Accumulibacter</i>	0.9922
Uncharacterised protein [†]	<i>A. baumannii</i> (WHO P1)	0.9995
GDSL-like Lipase [‡]	<i>A. baumannii</i> 625974 (WHO P1)	0.9993
Cyclolysin	<i>Candidatus Accumulibacter</i>	0.9810

Table 11: Zero-shot UniRef50 hits, ordered by novelty. *Ecp2: experimentally validated fungal effector—cross-kingdom generalisation from a probe trained exclusively on animal and bacterial toxins. [†]A0A009RXF9 (67 aa); AlphaFold pLDDT = 43.6 (fully disordered), candidate for wet-lab cytotoxicity assay. [‡]A0A009QBX0; annotated as “Uncharacterized protein” in UniProt, but confirmed via AlphaFold EBI structural homology as a GDSL-like Lipase/Acylhydrolase (pLDDT = 83.0, 44.6% residues > 90)—demonstrating the probe identifies functional threats that current databases have not yet annotated. Five organism groups account for 58% of all 248 hits: *C. Accumulibacter* (50), *Ruminococcus* (39), *Colletotrichum*/fungi (29), *Acinetobacter* incl. *baumannii* (26), and *Pseudomonas* (24).

3.12 The Probe Learned “Secreted Virulence,” Not “Toxin”

The zero-shot discovery above reveals something deeper than hit-finding: the probe encodes a representation *broader than its training label*. Three lines of evidence support this:

1. **Cross-kingdom transfer.** Ecp2 is an experimentally validated fungal effector that manipulates plant immune signalling—phylogenetically unrelated to any animal or bacterial toxin in the training set. Twenty-three total fungal hits from *Colletotrichum*/Pezizomycotina, most with signal peptides, confirm this is systematic, not coincidental.
2. **Structure-agnostic detection.** AlphaFold validation of all 248 hits shows an exactly 50/50 split between well-structured proteins (pLDDT ≥ 70) and intrinsically disordered proteins. Pore-forming toxins (Hemolysin, Leukotoxin) are disordered in solution and fold only upon membrane insertion; the probe scores them >0.99 despite low pLDDT. Conversely, GDSL Lipase (pLDDT = 83.0) and TonB receptor (pLDDT = 92.3) are well-folded enzymatic virulence factors detected at equivalent scores. The probe reads *secreted-virulence sequence motifs*, not fold topology.
3. **Signal peptide enrichment.** The 4.75 $\times$ enrichment for signal peptides among high-scoring hits (38% vs. 8% in low-scoring sequences) directly confirms the learned concept is “secreted”—the defining property of both toxins and effectors.

Deployment consideration. A 248/1,000 (24.8%) flag rate at the 0.85 threshold is too high for automated rejection in a synthesis-screening pipeline. However, the threshold is tunable: raising it to 0.95 reduces flags to 12.4% while retaining all confirmed virulence hits (Hemolysin, Leukotoxin, Ecp2, both *A. baumannii* proteins). Moreover, the 4.75 $\times$ signal peptide enrichment demonstrates that the majority of flags are functionally meaningful secreted proteins rather than noise, making them appropriate candidates for secondary review rather than false positives to be discarded. Critically, **135 of the 248 candidates (54.4%)** are currently annotated as “Uncharacterized protein” in UniProt, representing potential novel virulence factors that existing annotation pipelines have not identified. One such protein (A0A009QBX0, score 0.9993) was independently confirmed via AlphaFold structural homology as a GDSL-like Lipase/Acylhydrolase—a known class of secreted membrane-degrading virulence factors—illustrating the probe’s capacity for functional discovery ahead of conventional annotation.



Figure 8: UMAP projection of ESM-2 embeddings. Redesigns (red) cluster with natural toxins (orange), not controls (blue).

Representational Similarity Analysis (RSA):

Comparison	Mean cosine distance
Toxin $\leftrightarrow$ Toxin	0.0680
Toxin $\leftrightarrow$ Redesign	0.0647
Toxin $\leftrightarrow$ Control	0.1400
Redesign $\leftrightarrow$ Control	0.1453
Class separability	2.16$\times$
RSA Spearman r	-0.007 ($p = 0.648$)

Table 12: Redesigns are 2.16 $\times$ closer to toxins than to controls (0.065 vs 0.145), confirming ProteinMPNN preserves the toxin neighbourhood. However, internal pairwise geometry is not preserved (RSA $r = -0.007$)—ProteinMPNN scrambles relative positions while maintaining the class boundary.

4 Discussion

4.1 Why ProteinMPNN Fails to Evade ESM-2

ProteinMPNN optimises sequence identity reduction while preserving protein backbone geometry. Structure is precisely what ESM-2’s toxin circuit encodes. The transfer ratio of 1.28 is not merely “the probe still works”—it is evidence that ProteinMPNN redesign *amplifies* the structural motifs that make toxins detectable. Redesigns are 2.16 $\times$ closer to natural toxins than to controls in ESM-2 embedding space (RSA separability = 2.16), and the circuit correlation $r = 0.992$ confirms they activate the same representational pathway.

4.2 The Circuit Architecture

1. **Early layers (1–9):** Sequence-level features (AUROC 0.97 by L9)
2. **Mid layers (17–20):** Primary toxin discrimination (DPA +40–45)
3. **Suppressor layers (27–28):** Internal regulation
4. **Final layers (29–32):** Strong final-phase discrimination (DPA +23–130), with layer 32 as primary bottleneck
5. **Multiple pathways:** ProteinMPNN redesigns achieve the same endpoint via only 19% circuit overlap with natural toxins—ESM-2 has multiple routes to the toxin representation, making targeted disruption difficult

4.3 The Security Boundary

The four-tier attack taxonomy reveals the security boundary lies at gradient access:

- **No gradient + structural constraint (ProteinMPNN):** 6.1% evasion. Practical threat model—produces functional proteins via synthesis orders.
- **Gradient + no structural constraint (HotFlip):** 100% evasion. Requires white-box probe access; unconstrained substitutions may compromise protein function (to be confirmed by structural prediction).
- **Embedding-space attacks (PGD, pSSR/UAP):** 100% evasion. Require white-box access to ESM-2 internals; no sequence-space realisation without embedding inversion.

The UAP’s stable attack geometry ($\cos = -0.805$ invariant to ϵ) reveals why: the robust SAE features that bind all toxins together in embedding space (making them resistant to sequence attack) simultaneously create a thin manifold that one gradient-optimised vector can cross universally. **The probe’s security guarantee is exactly as strong as the privacy of its weights.**

4.4 Biosecurity Implications

1. **Sequence-based screening is insufficient:** All 723 redesigns evade BLAST.
2. **ESM-2 probes are robust by architecture:** Transfer ratio 1.28 means the adversary is working against themselves—ProteinMPNN preserves what the probe detects.
3. **No retraining needed:** 93.9% detection with zero exposure to redesigned sequences; cross-family holdout AUROC 0.9929 confirms generalisation to unseen toxin families.
4. **Zero-shot discovery:** The probe finds biosecurity threats not in any training set, including a WHO Priority 1 pathogen cluster and cross-kingdom fungal effectors—functioning as an open-ended discovery tool, not just a known-threat detector.
5. **Deployment recommendation:** Deploy ESM-2 probes with private weights. The security margin at $\epsilon = 0.5$ in embedding space is not practically reachable by synthesis-order adversaries without model internals.
6. **Interpretability enables targeted hardening:** The evadable features (transfer ratio <0.13) represent known blind spots. A screening system could flag anomalous patterns where robust features fire but evadable features do not—consistent with targeted evasion.

4.5 Limitations

- **HotFlip functional validation:** Whether HotFlip sequences retain toxin function is unconfirmed; structural constraints differ from ProteinMPNN’s explicit backbone preservation.
- **DMS single sequence:** The N29 polarity-not-identity finding is demonstrated on one representative toxin; generalisation across families requires additional DMS experiments.
- **Wet-lab validation:** Computational predictions require experimental confirmation, particularly the two uncharacterised *A. baumannii* candidates (0.9995, 0.9993).
- **Single redesign tool:** Extension to EvoDiff and RFDiffusion is ongoing.
- **Embedding inversion:** Embedding-space attacks do not directly produce sequences; the gap between embedding-space and sequence-space evasion quantifies the inversion cost.

5 Open Problems (Future Work)

1. **HotFlip transferability:** Do gradient-guided sequence attacks transfer across probe architectures and pLMs, analogous to GCG transferability across LLMs [2]? If so, probe diversity alone does not provide security.
2. **Protein alignment:** ProteinMPNN has no alignment objective—it will design dangerous proteins given the right backbone. Fine-tuning protein design models using circuit-level interventions at the layer-32 bottleneck (the protein-space analogue of RLHF) could make design models intrinsically safer rather than relying on output filters.
3. **Wet-lab validation:** AlphaFold structural prediction of the two uncharacterised *A. baumannii* candidates (UniRef50_A0A009RXF9, UniRef50_A0A009QBX0) followed by cytotoxicity assays would transform a computational prediction into a biosecurity discovery. Collaboration with IBBIS/CBH is the natural path.
4. **Attack taxonomy across pLMs:** A systematic benchmark of (pLM $\times$ probe architecture $\times$ attack type) would establish principled deployment recommendations—the protein-space analogue of AdvBench.

6 Conclusions

- **BLAST: 0%. ESM-2: 93.9%.** ProteinMPNN fails to evade structure-aware probes.
- **Transfer ratio 1.28:** Redesigns amplify toxin features—ProteinMPNN preserves what the probe detects.
- **$r = 0.992$:** Redesigns use the same DPA circuit as natural toxins; 19% circuit overlap means multiple routes to the same toxic endpoint.
- **Security boundary at gradient access:** 6.1% evasion (blackbox) vs 100% (white-box gradient). The probe’s guarantee equals the privacy of its weights.
- **UAP geometry:** Stable attack direction ($\cos = -0.805$, invariant to ϵ) reveals manifold structure, not probe weakness.
- **Zero-shot discovery:** Hemolysin, Leukotoxin, Cyclolysin, fungal effector Ecp2, and two uncharacterised *A. baumannii* proteins—none in training data.

- **Mutation robustness:** 0/1,179 single-point mutations evade; the circuit reads polarity, not sequence identity.

The key insight: ProteinMPNN cannot redesign away what is structurally necessary for toxin function—and ESM-2 reads structure.

Software Artifacts

1. **screen.py:** Zero-shot CLI screening tool with layer-wise DPA attribution.
2. **demo.ipynb:** Interactive notebook for screening and visualisation.

Key Numbers Reference

Metric	Value
Training toxins	1,712
Training controls	2,072
ProteinMPNN redesigns	723
Best probe layer	33
Probe AUROC (natural test)	0.9970
BLAST detection on redesigns	0.0%
ESM-2 detection on redesigns	93.9%
Double-Evaders (BLAST + Probe evasion)	92
SAE True Recovery of Double-Evaders	38% (35/92)
Mean transfer ratio	1.28
Cross-family holdout AUROC	0.9929
SAE features total	10,240
Dead SAE features	8,345 (81.5%)
Top-K features used	50
Compression ratio	205×
SAE AUROC (top-50)	0.9447
DPA tox/redesign correlation	$r = 0.992$
Circuit overlap (nat vs redesign)	19%
RSA class separability	2.16×
Steering ($\alpha = 2.0$, controls)	0.000 $\rightarrow$ 1.000
UAP security margin	$\epsilon = 0.5$
UAP cos (stable across all ϵ)	-0.805
DMS single-point evasion	0/1,179
UniRef50 candidates	248 / 1,000
Signal peptide enrichment	4.75× ($p < 0.001$)
ProteinMPNN evasion	6.1%
HotFlip evasion	100% [†]
PGD evasion	100%
pSSR evasion	100%

[†]HotFlip sequences may not retain protein function; wet-lab validation pending.

A Deep Double-Evader Scaffold Analysis

While the ESM-2 probe demonstrates a 93.9% overall detection rate against ProteinMPNN redesigns, the evasion rate is not uniformly distributed. We performed a scaffold-level analysis on the 89 “Double-Evaders” (sequences that evaded both BLAST and the ESM-2 probe).

A Kruskal-Wallis test reveals that evasion is highly scaffold-dependent. A small cluster of “susceptible” structural scaffolds consistently produce evading sequences (e.g., A0A348G5W2 and A0A835CKX4 at 100% evasion, P86523 at 86% evasion). Conversely, the vast majority of scaffolds (48 distinct fold families) are completely “robust”, producing 0% evasion across all their redesigns.

Furthermore, Double-Evaders are significantly further from training positives in embedding space than detected redesigns (mean L_2 distance 3.14 vs 2.49, Mann-Whitney $U = 20776$, $p < 0.001$). This indicates a *coverage problem* in the training distribution: the probe fails not because its architecture is weak, but because certain structural topologies were underrepresented during training, creating local blind spots in the decision manifold.

Susceptible Scaffolds (> 50% Evasion)	Rate	Representative Robust Scaffolds	Rate
A0A348G5W2	10/10 (100%)	P0DKN5	0/3 (0%)
A0A835CKX4	10/10 (100%)	P0DJE7	0/3 (0%)
P86523	6/7 (86%)	B1P1I0	0/10 (0%)
A0A6G9KJV6	8/10 (80%)	A0A7S8MVF3	0/6 (0%)
P0C8D4	7/9 (78%)	Q9PRQ3	0/10 (0%)
P0C7B1	7/10 (70%)	P24335	0/8 (0%)
P0DJB4	4/7 (57%)	P0DM71	0/4 (0%)

Table 13: Scaffold-level evasion rates. Evasion is highly bimodal: a few specific structural folds are systematically vulnerable to redesign-based evasion, while 48 other folds produce redesigns that are detected 100% of the time.

As demonstrated in Section 3.3, passing these Double-Evaders through a Sparse Autoencoder (SAE) successfully disentangles the concealed motifs, allowing a calibrated classifier to detect 38% of them. This confirms that even for highly susceptible scaffolds, the underlying toxic motifs are still retained in the latent representations, but require sparse extraction to overcome superposition.

References

- [1] Wittmann, B.J., et al. (2025). Strengthening nucleic acid biosecurity screening against generative protein design tools. *Science*.
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